MySQL - RDBMS

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SELECT – DQL – WHERE

- It is always good idea to fetch only required rows (to reduce network traffic).
- The WHERE clause is used to specify the condition, which records to be fetched.
- Relational operators
  - <, >, <=, >=, =, != or <>
- NULL related operators
  - NULL is special value and cannot be compared using relational operators.
  - IS NULL or <=>, IS NOT NULL.
- Logical operators
  - AND, OR, NOT

SELECT – DQL – WHERE

- BETWEEN operator (include both ends)
  - c1 BETWEEN val1 AND val2
- IN operator (equality check with multiple values)
  - c1 IN (val1, val2, val3)
- LIKE operator (similar strings)
  - c1 LIKE ‘pattern’.
  - % represent any number of any characters.
  - _ represent any single character.

UPDATE – DML

- To change one or more rows in a table.
- Update row(s) single column.
  - UPDATE table SET c2=new-value WHERE c1=some-value;
- Update multiple columns.
  - UPDATE table SET c2=new-value, c3=new-value WHERE c1=some-value;
- Update all rows single column.
  - UPDATE table SET c2=new-value;

UPDATE table SET name='Kapil' WHERE id=10;

DELETE vs TRUNCATE vs DROP

- **DELETE:** DML operation (can be rollbacked)
- Used to remove one or more rows from a table.
- Syntax:
  - DELETE FROM table WHERE c1 = value;
- Deletes specific rows that match the condition.
- Example:
  - DELETE FROM employees WHERE id = 10;

The row is marked as deleted.
It is not removed from the database

- Delete all rows:
- DELETE FROM temp;
- Removes all rows but keeps table structure.

- **TRUNCATE:** DDL operation (cannot be rollbacked)
- Used to delete all rows quickly.
- Syntax:
  - TRUNCATE TABLE table;
- Faster than DELETE.
- Cannot use WHERE with TRUNCATE.

Removes all rows but keeps table structure.

- **DROP :** DDL operation (cannot be rollbacked)
- Used to remove the table (and its structure) completely.
- Syntax:
  - DROP TABLE table;
- Deletes both data and table definition.
- Optional (avoid errors):
- DROP TABLE IF EXISTS table;
- Delete a whole database/schema:
  - DROP DATABASE db;

Seeking HELP

- HELP is client command to seek help on commands/functions.
    - HELP SELECT;
    - HELP Functions;
    - HELP SIGN;

SQL functions

- RDBMS provides many built-in functions to process the data.
- These functions can be classified as:
  - Single row functions
    - One row input produce one row output.
    - e.g. ABS(), CONCAT(), IFNULL(), …
  - Multi-row or Group functions
    - Values from multiple rows are aggregated to single value.
    - e.g. SUM(), MIN(), MAX(), …
- These functions can also be categorized based on data types or usage.
  - Numeric functions
  - String functions
  - Date and Time functions
  - Control flow functions
  - Information functions
  - Miscellaneous functions

Numeric & String functions

- ABS()
- POWER()
- ROUND(), FLOOR(), CEIL()

- ASCII(), CHAR()
- CONCAT()
- SUBSTRING()
- LOWER(), UPPER()
- TRIM(), LTRIM(), RTRIM()
- LPAD(), RPAD()

CHECK day03.sql file for detail.

Date-Time and Information functions

- VERSION()
- USER(), DATABASE()

- MySQL supports multiple date time related data types
  - DATE (3), TIME (3), DATETIME (5), TIMESTAMP (4), YEAR (1)
- SYSDATE(), NOW()
- DATE(), TIME()
- DAYOFMONTH(), MONTH(), YEAR(), HOUR(), MINUTE(), SECOND(), …
- DATEDIFF(), DATE_ADD(), TIMEDIFF()
- MAKEDATE(), MAKETIME()

Control and NULL and List functions

CHECK day03.sql for detailed explanation.

- NULL is special value in RDBMS that represents absence of value in that column.
- NULL values do not work with relational operators and need to use special operators.
- Most of functions return NULL if NULL value is passed as one of its argument.
- ISNULL()
- IFNULL()
- NULLIF()
- COALESCE()

- GREATEST(), LEAST()

- IF(condition, true-value, false-value)

THANK YOU